

CSA 0378-DATA STRUCTURES

25. BFS & DFS

main.c	Run	Output
<pre>1 #include <stdio.h> 2 int main() { 3 int graph[5][5] = { 4 {0,1,1,0,0}, 5 {1,0,0,1,1}, 6 {1,0,0,0,0}, 7 {0,1,0,0,0}, 8 {0,1,0,0,0} 9 }; 10 11 int queue[10], front=0, rear=0; 12 int visited[5]={0}; 13 int i, v; 14 15 visited[0]=1; 16 queue[rear++]=0; 17 18 printf("BFS: "); 19 while(front<rear) { 20 v=queue[front++]; 21 printf("%d ",v); 22 23 for(i=0;i<5;i++) { 24 if(graph[v][i] && !visited[i]) { 25 visited[i]=1; 26 queue[rear++]=i; 27 } 28 } 29 } 30 return 0; 31 }</pre>		BFS: 0 1 2 3 4 === Code Execution Successful ===

main.c	Run	Output
<pre>1 #include <stdio.h> 2 3 int graph[5][5] = { 4 {0,1,1,0,0}, 5 {1,0,0,1,1}, 6 {1,0,0,0,0}, 7 {0,1,0,0,0}, 8 {0,1,0,0,0} 9 }; 10 11 int visited[5]={0}; 12 13 void dfs(int v) { 14 int i; 15 16 visited[v]=1; 17 printf("%d ",v); 18 19 for(i=0;i<5;i++) 20 if(graph[v][i] && !visited[i]) 21 dfs(i); 22 } 23 24 int main() { 25 printf("DFS: "); 26 dfs(0); 27 return 0; 28 }</pre>		DFS: 0 1 3 4 2 === Code Execution Successful ===